# Lecture 18: Lagrangian Duality

## MA3236 Non-Linear Programming

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# Quick review: constrained optimization and KKT

L12–L14 studied constrained optimization theory.

$$\min_{\mathbf{x}\in\mathbf{R}^n} f(\mathbf{x}) \quad \text{s. t.} \quad \mathbf{g}(\mathbf{x}) = \mathbf{0}, \qquad \mathbf{h}(\mathbf{x}) \leq \mathbf{0},$$

where $\mathbf{g} : \mathbf{R}^n \to \mathbf{R}^m$, $\mathbf{h} : \mathbf{R}^n \to \mathbf{R}^p$.

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}) + \boldsymbol{\nu}^T \mathbf{h}(\mathbf{x}).$$

For differentiable data, the KKT conditions are

$$\begin{aligned}
\nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) &= \mathbf{0}, && \text{stationarity,} \\
\mathbf{g}(\mathbf{x}) = \mathbf{0}, \quad \mathbf{h}(\mathbf{x}) &\leq \mathbf{0}, && \text{primal feasibility,} \\
\boldsymbol{\nu} &\geq \mathbf{0}, && \text{dual feasibility,} \\
\nu_j h_j(\mathbf{x}) &= 0 \quad (\forall j), && \text{complementarity.}
\end{aligned}$$

# Quick review: primal methods

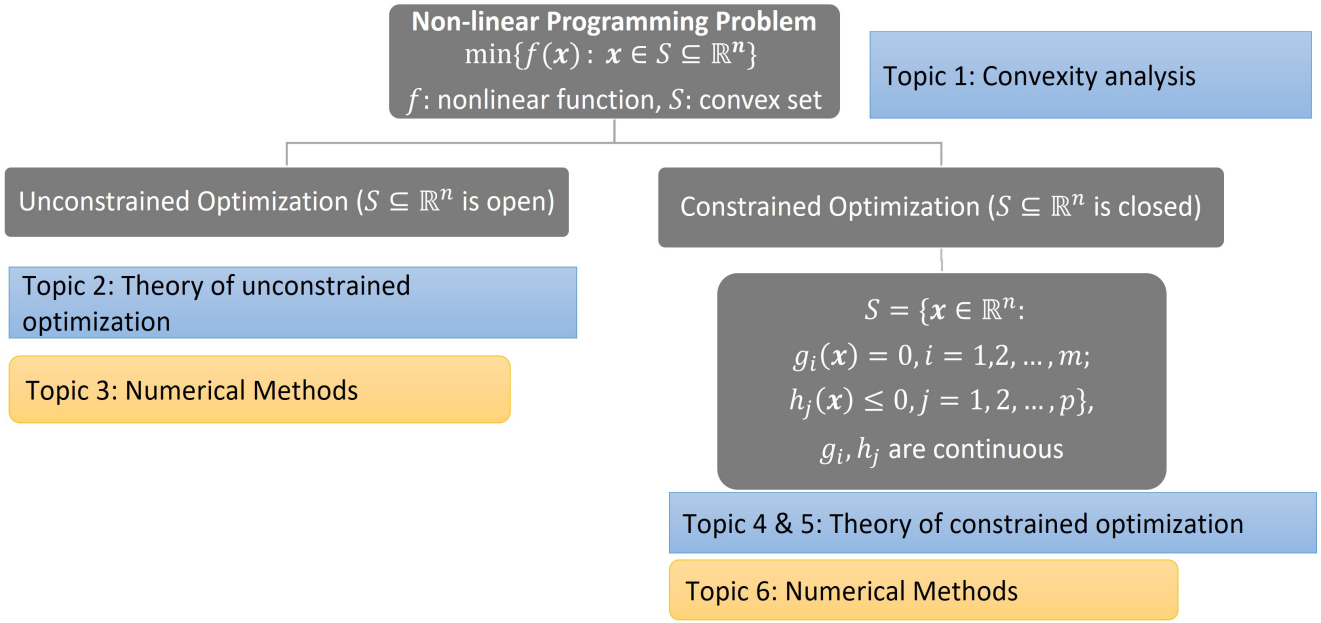
L15–L17 constructed subproblems to update $\mathbf{x}$.

| Method | Role of multipliers |
|---|---|
| QP | (Implicit) $\boldsymbol{\lambda}^{(k+1)} := \rho^{(k)}\mathbf{g}(\mathbf{x}^{(k+1)})$ |
| ALM | (Explicit) $\boldsymbol{\lambda}^{(k+1)} := \boldsymbol{\lambda}^{(k)} + \rho\mathbf{g}(\mathbf{x}^{(k+1)})$ |
| SQP | $\widehat{\boldsymbol{\lambda}}^{(k+1)}$: multiplier of the quadratic subproblem |
| Barrier | (Implicit) $\nu_j^{(k+1)} = -\mu^{(k)}/h_j(\mathbf{x}^{(k+1)})$ $(\forall j)$ |

- The primal variable $\mathbf{x}$ represents the answer to the original problem.
- The dual variables $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ are equally present in $L$.

What if we optimize over the multipliers themselves?

# Topics




In Topic 5, we study dual formulations and optimality theory for **constrained optimization problems**.

- Lagrangian dual and weak duality theorem.
- Strong duality theorem and saddle point optimality condition.

# Step 1: compare the Lagrangian with the primal objective

Take a primal feasible $\mathbf{x}$ and dual feasible $\boldsymbol{\nu} \geq \mathbf{0}$. Then

$$\mathbf{g}(\mathbf{x}) = \mathbf{0}, \qquad \mathbf{h}(\mathbf{x}) \leq \mathbf{0}, \qquad \nu_j h_j(\mathbf{x}) \leq 0,$$

so

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \cancel{\sum_{i=1}^{m} \lambda_i g_i(\mathbf{x})} + \sum_{j=1}^{p} \nu_j h_j(\mathbf{x})$$

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This bound still depends on both primal and dual variables.

# Step 2: eliminate the primal variable

For fixed multipliers, define the dual function

$$\boxed{\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\mathbf{z} \in \mathbf{R}^n} L(\mathbf{z}, \boldsymbol{\lambda}, \boldsymbol{\nu}).}$$

- The inf problem has **no constraints**.
- Its global infimum may not be attained; **it may equal** $-\infty$.

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- The inf problem has **no constraints**.
- Its global infimum may not be attained; **it may equal** $-\infty$.

For *every* primal feasible $\mathbf{x}$ and dual feasible $\boldsymbol{\nu} \geq \mathbf{0}$,

$$\boxed{\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq f(\mathbf{x}).}$$

One multiplier pair now gives a bound for all feasible primal points.

# Step 3: choose the best lower bound

$$\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq f(\mathbf{x}), \qquad \forall\ \mathbf{x} \in \mathcal{S}, \quad \boldsymbol{\nu} \geq \mathbf{0}.$$

$$\Downarrow$$

$$\boxed{\sup_{\boldsymbol{\lambda} \in \mathbf{R}^m, \boldsymbol{\nu} \geq \mathbf{0}} \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq \inf_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x})} \quad .$$

# Step 3: choose the best lower bound

$$\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq f(\mathbf{x}), \qquad \forall\, \mathbf{x} \in \mathcal{S}, \quad \boldsymbol{\nu} \geq \mathbf{0}.$$

$$\Downarrow$$

$$\boxed{d^* := \sup_{\boldsymbol{\lambda} \in \mathbf{R}^m, \boldsymbol{\nu} \geq \mathbf{0}} \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq \inf_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x}) =: p^*.}$$

**Theorem 18.1 (Weak duality)**

*The inequality $d^* \leq p^*$ holds, without convexity or regularity assumptions.*

- $d^*$: the supremum of Lagrangian lower bounds.
- $p^*$: the infimum of feasible primal objective values.

**NB.** With weak duality, nine cases $\rightarrow$ six cases.

# The best lower bound need not be exact

Consider the smooth problem

$$\min_{x \in \mathbf{R}} (x^2 - 1)^2 \quad \text{s. t.} \quad x = 0.$$

The only feasible point gives $p^* = 1$. Its Lagrangian is

$$L(x, \lambda) = (x^2 - 1)^2 + \lambda x.$$

We know $\theta(0) = 0$, and for every $\lambda$,

$$\theta(\lambda) \leq \min\{L(1, \lambda), L(-1, \lambda)\} = -|\lambda| \leq 0.$$

$$\boxed{d^* = 0 < 1 = p^*.}$$

L19: conditions for $d^* = p^*$ to hold (strong duality theorem).

# Duality gaps and an optimality certificate

For finite optimal values, the optimal duality gap is $\boxed{p^* - d^* \geq 0}$.

For primal feasible $\mathbf{x}$ and $\boldsymbol{\nu} \geq \mathbf{0}$, weak duality gives

$$\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq d^* \leq p^* \leq f(\mathbf{x}).$$

When $\theta(\boldsymbol{\lambda}, \boldsymbol{\nu})$ is finite, define the primal–dual gap

$$\boxed{G(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) - \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}),}$$

$$\Longrightarrow \quad 0 \leq \max\{p^* - d^*, f(\mathbf{x}) - p^*, d^* - \theta(\boldsymbol{\lambda}, \boldsymbol{\nu})\} \leq G.$$

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**Corollary 18.2 (Matching-value certificate)**

*If $G = 0$, then $\mathbf{x}$ and $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ are globally optimal.*

$G \leq \varepsilon_{\text{out}}$ certifies an objective error of at most $\varepsilon_{\text{out}}$, without derivatives.

# What looks attractive about the dual problem?

$$\sup_{\boldsymbol{\lambda}\in\mathbf{R}^m,\,\boldsymbol{\nu}\geq\mathbf{0}} \underbrace{\inf_{\mathbf{x}\in\mathbf{R}^n} \left\{ f(\mathbf{x}) + \boldsymbol{\lambda}^T\mathbf{g}(\mathbf{x}) + \boldsymbol{\nu}^T\mathbf{h}(\mathbf{x}) \right\}}_{\theta(\boldsymbol{\lambda},\boldsymbol{\nu})} .$$

1. Simple explicit constraints: only $\boldsymbol{\nu} \geq \mathbf{0}$ if $p > 0$.
2. Potential dimension reduction: $m + p$ multipliers versus $n$ variables.
3. Convex optimization: maximize a concave function on a convex set.

# Why is the dual function always concave?

**Theorem 18.3 (Concavity of the dual function)**

*$\theta$ is concave, even when the primal problem is not convex.*

Write $\mathbf{w} = (\boldsymbol{\lambda}, \boldsymbol{\nu})$. For fixed $\mathbf{x}$, $L(\mathbf{x}, \mathbf{w})$ is affine in $\mathbf{w}$. For $0 < t < 1$,

$$\begin{aligned}\theta(t\mathbf{w}_1 + (1-t)\mathbf{w}_2) &= \inf_{\mathbf{x}} \{tL(\mathbf{x}, \mathbf{w}_1) + (1-t)L(\mathbf{x}, \mathbf{w}_2)\} \\ &\geq t \inf_{\mathbf{x}} L(\mathbf{x}, \mathbf{w}_1) + (1-t) \inf_{\mathbf{x}} L(\mathbf{x}, \mathbf{w}_2) \\ &= t\theta(\mathbf{w}_1) + (1-t)\theta(\mathbf{w}_2).\end{aligned}$$

The inequality remains valid with values $-\infty$.

An infimum of affine functions is concave.

# A worked example

Consider

$$\min_{x_1, x_2 \in \mathbf{R}} \frac{1}{2}(x_1^2 + x_2^2) \quad \text{s. t.} \quad x_1 + x_2 = 4.$$

- Two primal variables, one equality multiplier $\lambda \in \mathbf{R}$.
- The constraint couples two otherwise independent costs.

We will construct the dual, maximize it, and recover a primal solution.

# Construct and evaluate the dual function

$$\min_{x_1,x_2\in\mathbf{R}} \frac{1}{2}(x_1^2+x_2^2) \quad \text{s. t.} \quad x_1+x_2=4.$$

Form

$$L(\mathbf{x},\lambda)=\tfrac{1}{2}x_1^2+\tfrac{1}{2}x_2^2+\lambda(x_1+x_2-4)$$

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For fixed $\lambda$, minimize each square independently:

$$x_1(\lambda)=x_2(\lambda)=-\lambda.$$

Therefore

$$\boxed{\theta(\lambda)=-\lambda^2-4\lambda=4-(\lambda+2)^2 \quad (\lambda\in\mathbf{R}).}$$

These inner responses $x_1(\lambda), x_2(\lambda)$ need not satisfy the original equality.

# Solve the dual and recover the primal solution

Maximize $\theta(\lambda) = 4 - (\lambda + 2)^2$:

$$\boxed{\lambda^* = -2, \qquad d^* = 4.}$$

The inner responses $x_1(\lambda) = x_2(\lambda) = -\lambda$ at this multiplier

# Solve the dual and recover the primal solution

Maximize $\theta(\lambda) = 4 - (\lambda + 2)^2$:

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The inner responses $x_1(\lambda) = x_2(\lambda) = -\lambda$ at this multiplier give

$$\mathbf{x}^* = (2, 2)^T, \qquad x_1^* + x_2^* = 4, \qquad f(\mathbf{x}^*) = 4.$$

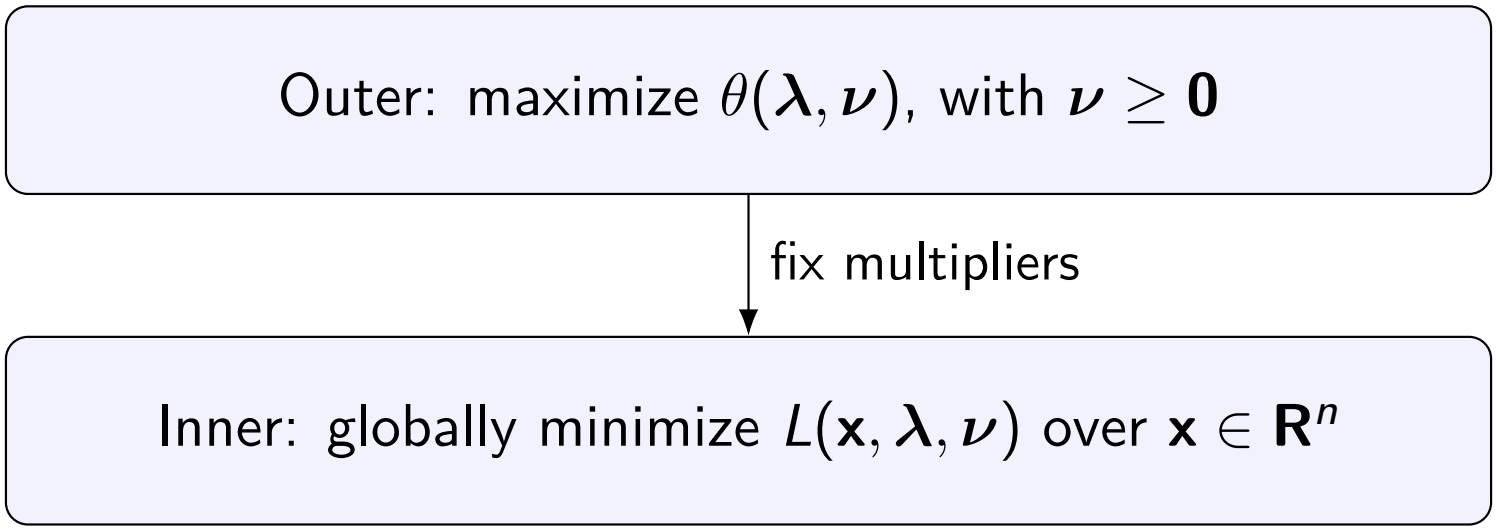
Weak duality now certifies

$$\boxed{4 = \theta(-2) \leq p^* \leq f(2, 2) = 4.}$$

Therefore, $\mathbf{x}^*$ and $\lambda^*$ are globally optimal solutions.

# The cost of the dual viewpoint: two optimization levels

Setting aside the duality gap...




- Inner problem: finite value? minimizer? global computation?
- Outer problem: finite supremum? maximizer? differentiability?
- Final goal: can we **recover a primal feasible optimum?**

# Inner inf: non-uniqueness, unattainability, unboundedness

$$\min_x \ e^x, \quad \text{s. t.} \quad 1 - e^x \leq 0.$$

The feasible set is $[0, \infty)$, so $x^* = 0$ and $p^* = 1$.

$$L(x, \nu) = (1 - \nu)e^x + \nu, \qquad \nu \geq 0.$$

| Multiplier | $\theta(\nu)$ | Inner behavior |
|---|---|---|
| $0 \leq \nu < 1$ | $\nu$ | Finite, not attained: $x \to -\infty$ |

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| $\nu = 1$ | 1 | Every $x \in \mathbf{R}$ minimizes $L(x, 1) = 1$ |
| $\nu > 1$ | $-\infty$ | **Unbounded below**: $x \to +\infty$ |

# The implicit effective domain

Define

$$\operatorname{dom}\theta = \{(\boldsymbol{\lambda}, \boldsymbol{\nu}) : \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) > -\infty\}.$$

The multipliers giving a finite lower bound lie in

$$\mathcal{F}_D = \operatorname{dom}\theta \cap (\mathbf{R}^m \times \mathbf{R}^p_+) \quad \rightsquigarrow \quad \sup_{(\boldsymbol{\lambda},\boldsymbol{\nu})\in\mathcal{F}_D} \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}).$$

For the example

$$L(x, \nu) = (1 - \nu)e^x + \nu, \qquad \nu \geq 0,$$

$\operatorname{dom}\theta = (-\infty, 1]$, but $\mathcal{F}_D = [0, 1]$.

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$$L(x, \nu) = (1 - \nu)e^x + \nu, \qquad \nu \geq 0,$$

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**NB.** Concavity makes $\operatorname{dom}\theta$, hence $\mathcal{F}_D$, convex.

# $\mathcal{F}_D$ can be empty with an unbounded primal

Suppose the primal problem is unbounded below: $p^* = -\infty$.

Weak duality: $\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) = -\infty$ for every $\boldsymbol{\lambda} \in \mathbf{R}^m, \boldsymbol{\nu} \in \mathbf{R}^p_+$.

$$\boxed{p^* = -\infty \implies d^* = -\infty, \quad \mathcal{F}_D = \emptyset.}$$

Can $\mathcal{F}_D$ be empty even when the primal is bounded below?

# $\mathcal{F}_D$ can be empty even with a bounded primal

$$\min_{x \in \mathbf{R}} -(x^2 - 1)^2 \quad \text{s. t.} \quad x = 0.$$

The only feasible point still exists, and now $\boxed{p^* = -1}$. However,

$$L(x, \lambda) = -(x^2 - 1)^2 + \lambda x \longrightarrow -\infty \quad (|x| \to \infty)$$

for every fixed $\lambda$. Hence

$$\boxed{\operatorname{dom} \theta = \mathcal{F}_D = \emptyset, \qquad d^* = -\infty.}$$

**NB.** A sufficient condition for $\mathcal{F}_D \neq \emptyset$:

$$\inf_{\mathbf{x} \in \mathbf{R}^n} f(\mathbf{x}) > -\infty.$$

# Completed primal-dual sudoku

| | $p^* = +\infty$ | $p^* \in \mathbf{R}$ | $p^* = -\infty$ |
|---|---|---|---|
| $d^* = +\infty$ | $\mathcal{S} = \emptyset$ | / | / |
| $d^* \in \mathbf{R}$ | $\mathcal{S} = \emptyset$ | $=$ or $<$ | / |
| $d^* = -\infty$ | $\mathcal{S} = \emptyset$ & $\mathcal{F}_D = \emptyset$ | $\mathcal{F}_D = \emptyset$ | $\mathcal{F}_D = \emptyset$ |

- Weak duality: nine cases $\rightarrow$ six cases.
- $p^* = +\infty$: primal infeasible; $d^* = -\infty$: no finite dual bound.
- $p^*, d^* \in \mathbf{R}$ can still have a positive optimal duality gap ($<$).

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- $p^*, d^* \in \mathbf{R}$ can still have a positive optimal duality gap ($<$).
- Besides finiteness, **uniqueness & attainability** are yet other issues.
- The outer objective $\theta(\boldsymbol{\lambda}, \boldsymbol{\nu})$ can also be **nondifferentiable**.

# Outer sup: non-uniqueness, unattainability, unboundedness

**Non-uniqueness**

Consider

$$\min_{x \in \mathbf{R}} x^2 \quad \text{s. t.} \quad x^2 \leq 0.$$

The primal solution is $x^* = 0$, and

$$L(x, \nu) = (1 + \nu)x^2, \qquad \nu \geq 0.$$

Every inner problem has the unique minimizer $x = 0$, so

$$\theta(\nu) = 0 \quad (\nu \geq 0), \qquad \boxed{d^* = 0, \quad \arg\max_{\nu \geq 0} \theta(\nu) = [0, \infty).}$$

# Outer sup: non-uniqueness, unattainability, unboundedness

**Unattainability**

$$\min_{x \in \mathbf{R}} \; x, \quad \text{s. t.} \quad x^2 \leq 0.$$

Again $p^* = 0$.

$$L(x, \nu) = x + \nu x^2 = \nu \left( x + \frac{1}{2\nu} \right)^2 - \frac{1}{4\nu} \quad (\nu > 0).$$

Hence

$$\theta(\nu) = \begin{cases} -\dfrac{1}{4\nu}, & \nu > 0, \\ -\infty, & \nu = 0. \end{cases}$$

$$\sup_{\nu \geq 0} \theta(\nu) = 0 = p^*, \qquad \text{no dual maximizer.}$$

# A smooth primal problem can have a nonsmooth dual

For

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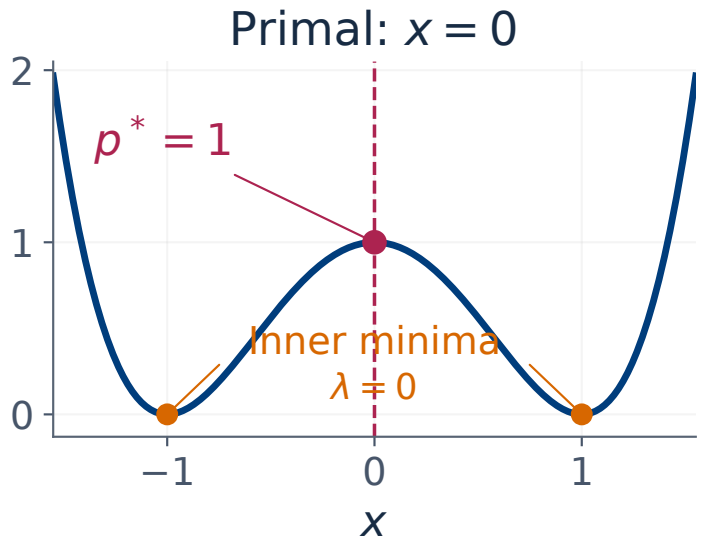
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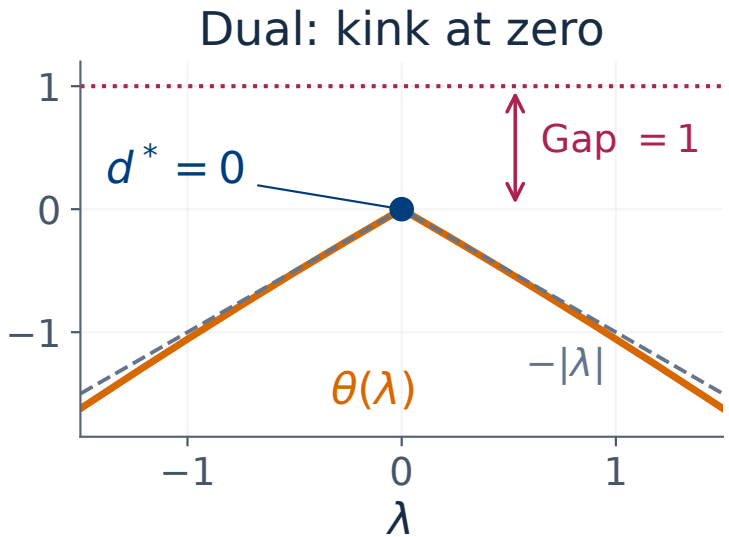
$$\theta(0) = 0, \quad x(0) \in \{-1, 1\}, \quad \theta(\lambda) \leq -|\lambda|.$$

Thus

$$\frac{\theta(\lambda) - \theta(0)}{\lambda} \begin{cases} \leq -1, & \lambda > 0, \\ \geq 1, & \lambda < 0. \end{cases}$$

A derivative at zero cannot exist!



# Primal recovery: choosing the right inner minimizer

Asume $-\infty < \theta(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = d^* = p^*$ and $\arg\min L(\cdot, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \neq \emptyset$.

$$\min_x \ e^x, \quad \text{s. t.} \quad 1 - e^x \leq 0.$$

The feasible set is $[0, \infty)$, so $x^* = 0$ and $p^* = 1$.

$$L(x, \nu) = (1 - \nu)e^x + \nu, \qquad \nu \geq 0.$$

At the dual optimum $\nu^* = 1$, every $x$ minimizes $L(x, 1) = 1$.

| Inner minimizer | Primal feasible? | Complementarity? | Primal optimal? |
|---|---|---|---|
| $x = -1$ | No | No | No |
| $x = 1$ | Yes | No | No |
| $x = 0$ | Yes | Yes | Yes |

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| $x = 1$ | Yes | No | No |
| $x = 0$ | Yes | Yes | Yes |

To recover a genuine primal optimum from an inner minimizer $\bar{\mathbf{x}}$,

$$\textbf{check} \quad \mathbf{g}(\bar{\mathbf{x}}) = \mathbf{0}, \quad \mathbf{h}(\bar{\mathbf{x}}) \leq \mathbf{0}, \quad \nu_j^* h_j(\bar{\mathbf{x}}) = 0.$$

# Approximate inner minimization can invalidate a bound

For a computed point $\widetilde{\mathbf{x}}$,

$$\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq L(\widetilde{\mathbf{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \not\leq p^*.$$

In the worked example, take $\lambda = -2$:

$$L(\mathbf{x}, -2) = 4 + \tfrac{1}{2}(x_1 - 2)^2 + \tfrac{1}{2}(x_2 - 2)^2.$$

At $\widetilde{\mathbf{x}} = (2 + e, 2)^T$, with any $e \neq 0$,

$$L(\widetilde{\mathbf{x}}, -2) = 4 + \tfrac{1}{2}e^2 > 4 = p^*.$$

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If a certified global error bound satisfies

$$L(\widetilde{\mathbf{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu}) - \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq \delta,$$

then

$$\boxed{L(\widetilde{\mathbf{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu}) - \delta \leq \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq p^*.}$$

# Take-home messages

**A dual-construction workflow.**

1. Form $L = f + \boldsymbol{\lambda}^T \mathbf{g} + \boldsymbol{\nu}^T \mathbf{h}$, with $\boldsymbol{\nu} \geq \mathbf{0}$.
2. Compute the global infimum in $\mathbf{x}$ over $\mathbf{R}^n$.
3. Identify the effective domain.
4. Maximize the possibly nonsmooth concave dual function.

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2. Compute the global infimum in $\mathbf{x}$ over $\mathbf{R}^n$.
3. Identify the effective domain.
4. Maximize the possibly nonsmooth concave dual function.
5. Check finiteness, attainability, uniqueness.
6. Recover a primal candidate; check feasibility and complementarity.
7. Check the duality gap.

# Take-home messages (cont'd)

| | Earlier primal methods | Dual viewpoint |
|---|---|---|
| Main construction | QP, ALM, SQP, barrier | $\inf_{\mathbf{z}} L(\mathbf{z}, \boldsymbol{\lambda}, \boldsymbol{\nu})$ |
| Main target | Approximate KKT | Best Lagrangian LB |
| Potential benefit | Direct progress on $\mathbf{x}$ | Simple explicit constraints, potentially fewer variables |

# Take-home messages (cont'd)

| | Earlier primal methods | Dual viewpoint |
|---|---|---|
| Main construction | QP, ALM, SQP, barrier | $\inf_{\mathbf{z}} L(\mathbf{z}, \boldsymbol{\lambda}, \boldsymbol{\nu})$ |
| Main target | Approximate KKT | Best Lagrangian LB |
| Potential benefit | Direct progress on $\mathbf{x}$ | Simple explicit constraints, potentially fewer variables |
| Main difficulty | Feasibility, conditioning, local convergence | Attainment, differentiability, primal recovery, inner global solve, possible gap |

The useful choice depends on the problem structure and the desired guarantee.